

Community Flood Monitoring and Reporting System

Project Proposal

Project Title

FloodWatch GH: A Community-Driven Flood Reporting System with Severity Prediction

Problem Statement

Flooding remains one of the most destructive recurring disasters in Ghana. Every rainy season, communities in Accra, Kumasi, Tamale and beyond face property damage, displacement, and loss of life. Yet when a flood occurs, the information available to residents and emergency responders is almost always incomplete, delayed, and unstructured.

People rely on WhatsApp forwards and Twitter posts to know which roads are impassable. NADMO responds after damage is already done. And critically, when a flood is reported in one area, nobody knows how far it is likely to spread — whether it will stay contained or push into neighbouring streets and communities.

That last question is what this project directly answers.

Project Overview

FloodWatch GH is a web-based platform that allows community members to report flood incidents in real time — with photos, descriptions, and location — while the system automatically analyses each report and generates a flood severity and spread prediction based on the terrain of the reported area.

The reporting side makes it a community awareness tool. The prediction side makes it an intelligence tool. Together, they create something more useful than either would be alone.

The Core Innovation — Severity Prediction

When a user submits a flood report with a location, the system does the following automatically in the background:

1. Extracts the coordinates of the reported incident
2. Queries elevation data for that location and its surrounding area using NASA's SRTM dataset (free and publicly available)
3. Analyses the terrain gradient — how steeply or gradually the land slopes away from the reported point
4. Estimates a spread zone — the likely area the floodwater could reach if the incident is not contained, based on how water naturally flows downhill across that terrain
5. Assigns a severity score — Low, Moderate, High, or Critical — and overlays the estimated spread zone on a map

This turns every community report into a predictive data point, not just a record.

How It Works — User Journey

For a resident: A person in Adenta notices their street is flooding. They open FloodWatch GH, drop a pin on their location, write a brief description, optionally attach a photo, and submit. Within seconds, the map updates with their report and shows an estimated spread zone around their area.

For a neighbour or commuter: Someone planning to travel through that area sees the report on the public dashboard, checks the severity score, and decides to take an alternate route.

For a local assembly or NADMO officer: They log into the admin dashboard using dedicated admin credentials and see a live map of active flood reports across their jurisdiction, colour-coded by severity, with spread estimates helping them prioritise which areas need immediate attention.

Technical Stack

Component	Technology
Frontend	React / Next.js 16 with TypeScript
Backend	Next.js API Routes (Node.js REST API)
Authentication	Cookie-based auth with role separation (Admin / Resident)
Database	PostgreSQL with PostGIS (Phase 2) — structured static data layer for prototype
Mapping	Leaflet.js
Elevation Data	SRTM via OpenTopography API
Spread Estimation	D8 Flow Direction Algorithm (TypeScript implementation)
File Storage	Cloudinary for photo uploads (Phase 2)
Hosting	Vercel + Railway

The Prediction Algorithm — Simplified

The severity prediction does not require a complex machine learning model. It uses a straightforward flow accumulation approach:

- Elevation data for the reported area is fetched as a raster grid from the OpenTopography API using NASA's SRTM dataset
- A basic D8 flow direction algorithm determines which direction water moves across each cell in the grid
- Cells that receive flow from the reported point are marked as the predicted spread zone
- The size and density of that zone determines the severity score (Low / Moderate / High / Critical)

This is academically grounded in hydrology and GIS literature, implemented in TypeScript within the Next.js API layer, and explainable to any examiner without requiring deep machine learning knowledge.

Development Phases

Phase 1 — Current (Prototype)

- Functional web application with user registration and flood incident reporting
- Cookie-based authentication with separate Admin and Resident roles
- Role-based access control — protected pages redirect unauthenticated users to login
- Admin-only dashboard accessible exclusively via dedicated admin credentials
- Automatic severity scoring and spread estimation using the D8 algorithm
- Interactive public map dashboard showing live reports colour-coded by severity
- Structured static data layer simulating real reports across five Greater Accra communities (Adenta, Accra New Town, Kumasi Adum, Tamale, Tema) as outlined in the project proposal
- Backend API routes structured and ready for database integration in Phase 2
- NADMO Dashboard shortcut in site footer accessible only to authenticated admin users

Phase 2 — Planned (Production)

- PostgreSQL with PostGIS database integration for persistent community report storage
- Prisma ORM connecting Next.js API routes to the live database without frontend changes
- Cloudinary integration for real photo uploads per flood report
- Real-time map updates as new community reports are submitted
- Full deployment on Vercel (frontend) and Railway (backend and database)

Scope and Deliverables

The project delivers the following within the academic timeline:

6. A functional web application with user registration and flood incident reporting
7. Role-based authentication separating regular residents from NADMO admin officers
8. Photo upload and geotagged location submission per report
9. Automatic elevation data fetching per reported location via OpenTopography API
10. A working severity scoring and spread estimation module using the D8 algorithm
11. An interactive public map dashboard showing live reports and predicted spread zones
12. A secure admin dashboard accessible only to NADMO officers via dedicated credentials
13. A pilot demonstration using simulated and real reports across five communities in Greater Accra

What Makes This Different

Feature	Basic Reporting Tool	FloodWatch GH
Community reporting	✓	✓
Photo uploads	✓	✓
Live map	✓	✓
Role-based authentication	✗	✓
Severity prediction	✗	✓

Spread zone estimation	✗	✓
Terrain-aware intelligence	✗	✓
Admin-only NADMO dashboard	✗	✓

The addition of role-based authentication, terrain-based severity prediction, and an admin-only NADMO dashboard separates FloodWatch GH from a straightforward CRUD application and positions it as a decision-support tool with real-world utility.

Expected Impact

- Residents make faster, informed decisions about evacuation or alternate routes
- Emergency responders gain a prioritised view of where intervention is most urgent
- Local assemblies build a growing database of flood incident history mapped to terrain
- The severity model creates a foundation that can be improved with real historical flood data over time
- NADMO officers can access a dedicated secure dashboard to manage and respond to active flood reports

Admin Access and Security

FloodWatch GH implements a two-tier access system:

- Regular residents register with any email and password to submit and view flood reports
- NADMO admin officers use dedicated admin credentials to access the protected admin dashboard
- All protected routes are enforced at the middleware level — unauthenticated users are automatically redirected to the login page
- Regular users who attempt to access the admin dashboard are redirected to the homepage
- The NADMO Dashboard link in the site footer serves as an admin shortcut — it redirects non-admin users to the login page before granting access

Repository

GitHub: <https://github.com/Brefo205/FloodWatch-GH>